MODULE 4A · EPISODE 2

Kepler's Laws and Their Applications

From Tycho's Mars Data to Weighing the Stars

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Kepler's Laws and Their Applications

This episode turns to Johannes Kepler, the man who made the heliocentric model work. Using Tycho Brahe's careful observations of Mars, Kepler found three laws of planetary motion: orbits are ellipses with the Sun at one focus, a planet sweeps out equal areas in equal times, and the square of a planet's year is proportional to the cube of its average distance from the Sun. The panel shows how the Second Law shapes Earth's own seasons, and how the Third Law, combined with the transits of Venus, revealed the size of the entire Solar System. It then puts gravity to work in a pair of thought experiments, follows the Moon from its violent birth to its present distance and its locked rotation, and closes with the Third Law's most powerful modern use: measuring the masses of stars in binary systems many light-years away.

The Panel

Name	Role
Dr. Edward Thorne	Host; astrophysicist
Dr. Lena Petrova	Guest; exoplanet researcher
Dr. Kenji Tanaka	Guest; microbial ecologist

Learning Objectives

After this episode you should be able to:

- Explain why Kepler holds a unique place in astronomy, and describe the role of Tycho Brahe's observations.
- State Kepler's First Law and explain how it corrected Copernicus's model.
- State Kepler's Second Law and explain why a planet moves fastest when it is closest to the Sun.
- Describe how the Second Law affects Earth's daylight and the length of its seasons.
- State Kepler's Third Law and explain the meaning of orbital period and semi-major axis.
- Describe how Newton's laws provided the theoretical foundation for Kepler's Laws.
- Explain how the transits of Venus measured the astronomical unit, and how the Third Law then gave the size of the Solar System.
- Use Newton's gravity and his Third Law to explain why everyday gravitational forces are tiny and why the Earth doesn't rise to meet you when you jump.
- Describe the Moon's formation, its outward migration, the lengthening of Earth's day, and tidal locking.
- Explain how Kepler's Third Law, eclipses, and Doppler shifts reveal the masses of binary stars.

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Key Terms

Term	Definition
Kepler's First Law	The orbit of a planet is an ellipse, with the Sun at one of the ellipse's two foci.
Ellipse	An oval curve, a “squished circle,” with two foci.
Focus (plural: foci)	One of the two fixed points inside an ellipse. The Sun sits at one focus of a planet's orbit; the other focus is empty.
Kepler's Second Law	A line joining a planet and the Sun sweeps out equal areas during equal intervals of time.
Kepler's Third Law	The square of a planet's orbital period is directly proportional to the cube of the semi-major axis of its orbit.
Orbital period	The time a planet takes to complete one orbit; the planet's year.
Semi-major axis	Half the longest diameter of an ellipse; for a planet, its average distance from the Sun.
Solstice	One of the two times each year, in June and December, when the Sun is farthest north or south in the sky.
Transit of Venus	A rare event in which Venus passes across the face of the Sun as seen from Earth.
Astronomical unit (au)	The average distance between the Earth and the Sun, about 93 million miles.
Newton's Law of Universal Gravitation	Every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between their centers.
Newton's Third Law of Motion	For every action, there is an equal and opposite reaction.
Tidal forces	Forces that come from the difference in gravity's pull across a body. They pushed the Moon outward and slowed its rotation.
Angular momentum	The “rotational momentum” of a spinning or orbiting body. Earth has transferred some of its spin angular momentum to the Moon.
Tidal locking	The state in which a body rotates on its axis exactly once for every revolution it makes around its partner, so it always shows the same face.
Near side	The side of the Moon that always faces Earth.
Binary star system	Two stars orbiting each other. Such systems are very common.
Eclipsing binary	A binary star system in which the stars pass in front of each other as seen from Earth, causing dips in the system's total brightness.
Doppler shift	The change in the wavelength of light from a moving source: shorter (bluer) when the source moves toward us, longer (redder) when it moves away.

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Term	Definition
Light-year	The distance light travels in one year, about 6 trillion miles.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
Early January	When Earth is closest to the Sun, and moving fastest in its orbit
~4 days	How much shorter Northern Hemisphere winter is than summer, the longest season
1761, 1769	The transits of Venus used to measure the au
~95 million mi	Estimate of the au from the combined transit observations
93 million mi	Modern value of the au
80 kg, 1,589 kg, 1 m	The person and car in the gravity thought experiment
8.5×10^{-6} N	Their gravitational attraction, about the weight of a small worker ant
1.3×10^{-22} m/s²	The acceleration an 80 kg person gives the Earth
~4 billionths	How far the Earth rises, as a fraction of a hydrogen nucleus's diameter, when you jump half a meter
20,000–30,000 km	The Moon's distance from Earth when it formed
384,000 km	The Moon's average distance today
4.5 billion years	The time over which tidal forces moved the Moon outward
4–6 h → 24 h	The length of Earth's day then and now
~1 Earth month	The length of one Moon day

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — Kepler's Laws of Planetary Motion

Johannes Kepler holds a unique and revered place in the history of astronomy. He paved the way for the acceptance of the heliocentric model by “fixing” the Copernican model of the Solar System, and his laws came not from theory, but from careful observational data.

A contemporary of Galileo, Kepler worked for **Tycho Brahe**, the last great naked-eye astronomer. When Brahe died, Kepler inherited—some would say took—his priceless observations of Mars, which allowed him to work out his three laws of planetary motion.

Kepler's First Law states that the orbit of a planet is an **ellipse**, with the Sun at one of the ellipse's two **foci**. Think of an orbit as a squished circle. This was the key insight that corrected Copernicus's erroneous assumption of perfectly circular orbits.

Part 2 — The Second and Third Laws

Kepler's Second Law states that a line segment joining a planet and the Sun sweeps out **equal areas during equal intervals of time**. So when a planet is closer to the Sun, it must move faster.

This affects us directly. Earth is closest to the Sun in **early January**, so it moves fastest around the December solstice. That is why the day-to-day changes in daylight around the December solstice are slightly larger than those around the June solstice, and why Northern Hemisphere **winter is the shortest season**, by about four days.

Kepler's Third Law, perhaps the most valuable to astronomers, states that the square of a planet's **orbital period** is directly proportional to the cube of the **semi-major axis** of its orbit. In simpler terms, the square of a planet's year is proportional to the cube of its average distance from the Sun, a precise mathematical link between a planet's period and its distance.

Newton was later able to **derive all three of Kepler's Laws** from his own laws of motion and his law of gravity, providing the theoretical foundation that Kepler's work was missing.

Part 3 — Measuring the Solar System

Newton's friend **Edmond Halley** proposed that the **transits of Venus** in 1761 and 1769—rare events in which Venus passes across the face of the Sun—could be used to find the average distance between the Sun and the Earth, the unit we now call the **astronomical unit**, or **au**. Sadly, Halley didn't live to see either transit.

Numerous groups were sent to different locations on Earth to observe the transits. Their combined results gave an estimate of about **95 million miles**, remarkably close to the modern value of **93 million miles**.

Once the au was known, astronomers could use **Kepler's Third Law** to calculate the average distance to all the other planets, and so determine the size of the entire Solar System.

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Part 4 — Gravity in Action

How attractive are you, gravitationally speaking? The gravitational attraction between an **80-kilogram person** and a **1,589-kilogram car** standing one meter apart is about **8.5×10^{-6} newtons**, roughly the weight of a small worker ant. It's a tiny force.

When you jump, the Earth pulls you down, but by **Newton's Third Law** you also pull the Earth up with an equal force. The Earth doesn't come up to meet you because, although the forces are equal, the **accelerations are not**: the Earth's mass is enormous compared with yours. An 80-kilogram person gives the Earth an acceleration of only about **1.3×10^{-22} m/s²**. Jump half a meter high, and the Earth rises to meet you by only about **four billionths of the diameter of a hydrogen nucleus**.

Part 5 — The Earth-Moon System

The Moon formed from debris thrown off when a **Mars-sized object struck the early Earth**. At first it was only about **20,000 to 30,000 kilometers** away. Over 4.5 billion years, **tidal forces** have moved it outward to its current average distance of about **384,000 kilometers**.

The same process transferred some of Earth's rotational momentum, its **angular momentum**, to the Moon, lengthening our day from about **4 to 6 hours** to the **24 hours** we have now.

Early in its history, Earth's gravity slowed the Moon's rotation until it turned on its axis exactly once for every revolution around the Earth. This is **tidal locking**. As a result, one Moon day is about one Earth month long, and the same side, the **near side**, always faces us.

Part 6 — Kepler's Third Law in Modern Astronomy

Kepler's Third Law applies to **any two objects orbiting each other**, including **binary star systems**, which are very common. If we can measure both the period of the orbit and its size, the law gives the two stars' **combined mass**.

Eclipsing binaries are especially useful. When one star passes in front of the other, the system's total brightness dims, and timing those dips gives the **orbital period**. Combine that with the stars' orbital speeds, measured from **Doppler shifts** in their light, and we can work out the size of the orbit and even the **mass of each individual star**, a powerful tool for understanding stars many light-years away.

From elliptical orbits to the masses of distant stars, Kepler's Laws remain a cornerstone of astronomy.

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STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Kepler's Laws of Planetary Motion

1. Why does Kepler hold a unique place in the history of astronomy? Where did his laws come from?

ANSWER He “fixed” the Copernican model of the Solar System and paved the way for the acceptance of the heliocentric model. His laws came not from theory but from careful observational data.

2. Who was Tycho Brahe, and what did Kepler gain from him?

ANSWER Tycho Brahe was the last great naked-eye astronomer, and Kepler worked for him. When Brahe died, Kepler inherited (some would say took) his priceless observations of Mars, which allowed him to work out his three laws.

3. State Kepler's First Law of Planetary Motion.

ANSWER The orbit of a planet is an ellipse, with the Sun at one of the ellipse's two foci.

4. What is an ellipse? Where is the Sun located in a planet's orbit, and what is at the other focus?

ANSWER An ellipse is an oval curve, a squished circle, with two foci. The Sun sits at one focus; the other focus is empty.

5. Which assumption in Copernicus's model did Kepler's First Law correct?

ANSWER Copernicus assumed that planets move in perfectly circular orbits. The First Law showed that orbits are ellipses.

Part 2 — The Second and Third Laws

1. State Kepler's Second Law of Planetary Motion.

ANSWER A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.

2. What does the Second Law tell us about a planet's speed? Where in its orbit is a planet fastest?

ANSWER A planet's speed changes along its orbit. It moves fastest when it is closest to the Sun and slowest when it is farthest away.

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3. When is Earth closest to the Sun, and how does this affect Earth's motion around the December solstice?

ANSWER In early January. Because Earth is near its closest point, it moves fastest along its orbit around the December solstice.

4. Describe two effects on Earth of its faster orbital motion around the December solstice.

ANSWER The day-to-day changes in daylight around the December solstice are slightly larger than those around the June solstice, and Northern Hemisphere winter is the shortest season of the year, by about four days.

5. State Kepler's Third Law, first formally and then in simpler terms.

ANSWER The square of the orbital period of a planet is directly proportional to the cube of the semi-major axis of its orbit. In simpler terms, the square of a planet's year is proportional to the cube of its average distance from the Sun.

6. When a planet's year is measured in Earth years and its distance in au, the Third Law becomes $T^2 = a^3$. A planet's average distance from the Sun is 4 au. How long is its year?

ANSWER $T^2 = 4^3 = 64$, so T = 8 years.

7. What did Newton add to Kepler's Laws?

ANSWER Newton derived all three of Kepler's Laws from his own laws of motion and his law of gravity, providing the theoretical foundation that Kepler's work was missing.

Part 3 — Measuring the Solar System

1. Who proposed using the transits of Venus to measure the Solar System, and what were they meant to determine?

ANSWER Edmond Halley, Newton's friend. The transits could be used to determine the average distance between the Sun and the Earth.

2. What is a transit of Venus? In which years did the transits Halley proposed occur, and did he see them?

ANSWER A rare event in which Venus passes across the face of the Sun as seen from Earth. They occurred in 1761 and 1769. Halley didn't live to see either one.

3. How were the transits observed, and how did the result compare with the modern value?

ANSWER Numerous groups were sent to different locations on Earth to observe them. Their combined results gave about 95 million miles, remarkably close to the modern value of 93 million miles.

4. What is the astronomical unit?

ANSWER The average distance between the Sun and the Earth, about 93 million miles, abbreviated au.

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5. Once the au was known, how did astronomers determine the size of the entire Solar System?

ANSWER They used Kepler's Third Law to calculate the average distance from the Sun to each of the other planets.

Part 4 — Gravity in Action

1. How large is the gravitational force between an 80-kilogram person and a 1,589-kilogram car standing one meter apart? What is it comparable to?

ANSWER About 8.5×10^{-6} newtons, roughly the weight of a small worker ant.

2. Why is the gravitational attraction between everyday objects, like a person and a car, so tiny?

ANSWER Gravity is an extremely weak force. It becomes large only when at least one of the masses is enormous, like a planet.

3. When you jump, the Earth pulls you down. According to Newton's Third Law, what do you do to the Earth?

ANSWER You pull the Earth up with a force equal in size and opposite in direction to the Earth's pull on you.

4. If the forces are equal, why doesn't the Earth come up to meet you?

ANSWER The forces are equal, but the accelerations are not. The Earth's mass is enormous compared with yours, so the acceleration you give it is incredibly small, about 1.3×10^{-22} m/s² for an 80-kilogram person.

5. If you jump half a meter high, about how far does the Earth rise to meet you?

ANSWER Only about four billionths of the diameter of a hydrogen nucleus, an infinitesimally small distance.

Part 5 — The Earth-Moon System

1. How did the Moon form?

ANSWER From debris thrown off when a Mars-sized object struck the early Earth.

2. How far from Earth was the Moon when it formed, and how far is it now? What caused the change?

ANSWER About 20,000 to 30,000 kilometers at first, and about 384,000 kilometers on average today. Tidal forces have moved it outward over 4.5 billion years.

3. What did Earth transfer to the Moon as the Moon moved outward, and how did this change the length of Earth's day?

ANSWER Some of its rotational momentum (angular momentum). Earth's day lengthened from about 4 to 6 hours to the 24 hours we have now.

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4. What is tidal locking, and how did it happen to the Moon?

ANSWER Tidal locking is when a body rotates on its axis exactly once for every revolution it makes around its partner. Early in the Moon's history, Earth's gravity slowed the Moon's rotation until it reached that state.

5. Why do we always see the same side of the Moon? How long is one day on the Moon?

ANSWER Because the Moon is tidally locked, the same side, the near side, always faces Earth. One Moon day is about one Earth month long.

Part 6 — Kepler's Third Law in Modern Astronomy

1. To which systems besides planets orbiting the Sun can Kepler's Third Law be applied? Give an example.

ANSWER To any two objects orbiting each other, for example binary star systems, which are very common.

2. Which two measurements of a binary star's orbit let us find the stars' combined mass?

ANSWER The period of the orbit and its size.

3. What is an eclipsing binary, and how does it reveal its orbital period?

ANSWER A binary in which one star passes in front of the other as seen from Earth. Each eclipse dims the system's total brightness, and timing those dips gives the orbital period.

4. How are the stars' orbital speeds measured, and what do they add to the period?

ANSWER From Doppler shifts in the stars' light. Combined with the period, the speeds give the size of the orbit and even the mass of each individual star.

5. The episode calls Kepler's Laws “a cornerstone of astronomy.” Use examples from the episode to explain why.

ANSWER They describe how planets move (elliptical orbits, faster near the Sun) and explain effects on Earth such as the shorter Northern Hemisphere winter. The Third Law, scaled by the au from the transits of Venus, gave the size of the Solar System, and today it measures the masses of stars many light-years away.

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TEACHER NOTES

Corrections and Cautions

Points in the episode that need a qualification, or a prepared answer for the question a sharp student will ask.

C1 **Laws from data, explanation from Newton.**

Kepler's Laws are empirical: he found them by fitting Tycho's observations. He published the First and Second Laws in 1609 and the Third in 1619. Newton explained why they hold in his Principia (1687), and his version extends them to any two orbiting bodies.

C2 **“Inherited—some would say took.”**

Tycho died in 1601, and his heirs claimed his observations. Kepler kept the Mars records and used them for years. It's a good moment to discuss how much science depends on access to good data.

C3 **The ellipses are only slightly squished.**

Most planetary orbits are nearly circular: Earth's eccentricity is 0.017 and Mars's is 0.093. Diagrams, including the episode's, exaggerate the ovals. Kepler chose Mars because its orbit is one of the more eccentric, and his best circular fit still missed Tycho's positions by 8 arcminutes, about a quarter of the Moon's width. The second focus is empty; nothing sits there.

C4 **Closest in January, yet it's winter.**

Students may conclude that Earth's distance causes the seasons. It doesn't: the seasons come from the tilt of Earth's axis, and Northern Hemisphere winter happens while Earth is closest to the Sun. The distance changes only the seasons' lengths.

C5 **“Shortest by about four days.”**

Northern Hemisphere winter lasts about 89.0 days and summer, the longest season, about 93.7 days, so the four-plus-day gap is measured against summer. Autumn is less than a day longer than winter. In the Southern Hemisphere the pattern flips: the short season is summer.

C6 **The daylight effect is real but small.**

Faster motion near the December solstice makes the Sun's north-south position change a little faster, so day-to-day daylight changes are slightly larger then. A more noticeable consequence of the same faster motion is that the earliest sunsets come about two weeks before the December solstice and the latest sunrises about two weeks after it.

C7 **Two forms of the Third Law.**

For bodies orbiting the Sun, with T in years and a in au, the law becomes $T^2 = a^3$. Newton's version, $T^2 = 4\pi^2 a^3 / G(M_1 + M_2)$, includes the masses, which is what lets Part 6 weigh binary stars. The semi-major axis is the average of a planet's closest and farthest distances; calling it the average distance is accurate for nearly circular orbits.

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C8 **The transits of Venus.**

Halley proposed the method in 1716 and died in 1742. Transits come in pairs eight years apart, separated by more than a century: 1761 and 1769, 1874 and 1882, 2004 and 2012; the next are in 2117 and 2125. Observers far apart see Venus cross the Sun along slightly different paths, a parallax that gives the distance. The 95-million-mile figure comes from Johann Encke's 1824 analysis of the 1761 and 1769 data. The au is now fixed at exactly 149,597,870.7 km, about 92.96 million miles.

C9 **Relative distances came first.**

Even before the au was known, the Third Law gave every planet's distance in au from its period alone: a complete map of the Solar System without a scale. The transits supplied the scale.

C10 **The Gravity in Action numbers.**

$F = Gm_1m_2/r^2 = (6.674 \times 10^{-11})(80)(1{,}589)/1^2 \approx 8.5 \times 10^{-6}$ N, the weight of about 0.9 mg, a small ant. This treats the person and car as point masses with centers 1 m apart. Earth's acceleration is 784 N ÷ 5.97×10^{24} kg ≈ 1.3×10^{-22} m/s², and its rise during a 0.5 m jump is 0.5 m × 80/(5.97×10^{24}) ≈ 7×10^{-24} m, about 4 billionths of a proton's 1.7×10^{-15} m. Strictly, both bodies move around their common center of mass.

C11 **The giant impact is the leading idea, not a certainty.**

The impactor is often called Theia. The giant-impact hypothesis best explains the Moon's composition and Earth's spin, but the details are still debated, and estimates of the Moon's starting distance vary from model to model. The Moon is still receding, about 3.8 cm per year, measured by bouncing lasers off mirrors left by Apollo astronauts.

C12 **“Rotational momentum” means angular momentum.**

Earth's spin drags the Moon's tidal bulges slightly ahead of the Earth-Moon line. The nearer bulge tugs the Moon forward, lifting it into a wider orbit, while the Moon's pull back on the bulges slows Earth's spin. Total angular momentum is conserved. Running the transfer backward, with the Moon at 20,000–30,000 km, gives a day of about 5 hours, in line with the episode. Today the day lengthens by roughly 2 milliseconds per century. Surprisingly, the Moon moves more slowly as it moves outward.

C13 **Tidal locking details.**

The Moon rotates once every 27.3 days relative to the stars; its day from one sunrise to the next is 29.5 days, which is the “about one Earth month.” Because of slight wobbles (librations), we see about 59% of the Moon's surface over time. The far side is not a “dark side”: it gets as much sunlight as the near side.

C14 **Weighing binary stars.**

Roughly half of Sun-like stars have one or more companions. In Newton's form, with masses in solar masses, $M_1 + M_2 = a^3/P^2$. Finding a binary's size from pictures requires knowing its distance; eclipsing binaries avoid that. Their orbits are nearly edge-on, so the Doppler speeds are close to the true orbital speeds: $a = (v_1 + v_2)P/2\pi$, and $M_1/M_2 = v_2/v_1$. The heavier star moves more slowly.

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TEACHER NOTES

Teaching the Episode

Draw the orbits.

Run the pins-and-string activity from the Try It box. Have students mark the Sun at one pin, then measure where their “planet” is closest and farthest. Ask why the Second Law says it moves fastest at the closest point.

Check the Third Law with real planets.

Give students each planet's distance and year: Mercury 0.387 au and 0.241 years, Venus 0.723 and 0.615, Earth 1 and 1, Mars 1.524 and 1.881, Jupiter 5.203 and 11.86, Saturn 9.537 and 29.46. Have them compute T^2 and a^3 for each. The two columns match, the same discovery Kepler made in 1619.

Count the days in the seasons.

Using a calendar, have students count from the December solstice to the March equinox, and from the June solstice to the September equinox. Winter comes out about four or five days shorter. Then ask what that says about Earth's speed in January.

See parallax for yourself.

Have students hold up a thumb at arm's length and look at it with one eye, then the other. The thumb jumps against the background, just as Venus's path across the Sun shifted for observers in different places. Photos of the 2012 transit make the event concrete.

Work the thought experiments on the board.

Compute the person-car force together, then the Earth's acceleration from a jump. Ask students to explain, in one sentence, why equal forces can give such unequal results.

Walk around the chair.

Have a student walk in a circle around a chair while always facing it. Ask how many times the student turned around in one lap. The answer, once, is tidal locking.

Read a light curve.

Sketch an eclipsing binary's brightness over time, with a deep dip and a shallow one each orbit, and have students mark one period. The famous eclipsing binary Algol, in Perseus, dims noticeably to the naked eye every 2.87 days.

Use the panel.

Kenji brings an ecologist's angle. When the young Moon was ten or more times closer, tides were enormously stronger. Ask students how today's tide pools, home to some of Earth's richest communities of life, depend on the Moon's present distance.

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